

QUINOFORT 100MG/ML INJECTION

Description and Composition

QUINOFORT 100MG/ML INJECTION is a light yellow to yellow clear solution. Each ml contains 100mg of Enrofloxacin as its active ingredient and 20mg Benzyl Alcohol as its preservative.

Pharmacodynamics

Enrofloxacin has a spectrum of activity which includes enrofloxacin-sensitive *Histophilus somni*, *Mannheimia haemolytica*, *Pasteurella multocida*, *Mycoplasma* spp., and *E. coli* in cattle as well as *Actinobacillus pleuropneumoniae*, *Pasteurella multocida* and *Haemophilus parasuis* in pigs.

Enrofloxacin belongs to the fluoroquinolone group of antibiotics. The substance has bactericidal activity which is mediated by binding to subunit A of DNA gyrase and the resulting selective inhibition of this enzyme.

DNA gyrase is a topoisomerase. These enzymes are involved in the replication, transcription and recombination of bacterial DNA. Fluoroquinolones also influence bacteria in the stationary phase by altering cell wall permeability.

Resistance to fluoroquinolones has been reported to arise from five sources, (i) point mutations in the genes encoding for DNA gyrase and/or topoisomerase IV leading to alterations of the respective enzyme, (ii) alterations of drug permeability in Gram-negative bacteria, (iii) efflux mechanisms, (iv) plasmid mediated resistance and (v) gyrase protecting proteins. All mechanisms lead to a reduced susceptibility of the bacteria to fluoroquinolones. Cross-resistance within the fluoroquinolone class of antimicrobials is common.

The inhibitory and bactericidal concentrations of enrofloxacin are very close, being either identical or differing by no more than 1-2 dilution steps.

Pharmacokinetics

Following subcutaneous administration of the product in cattle or intramuscular administration in pigs, the active substance, enrofloxacin, is absorbed very rapidly and almost completely (high bioavailability).

Cattle:

After subcutaneous administration at a dose rate of 7.5 mg enrofloxacin per kg body weight to non-lactating cattle peak plasma concentrations of 0.82 mg/L are reached within 5 hours. The overall drug exposure in plasma is 9.1 mg*hr/L. Enrofloxacin is eliminated from the body at a half-life of 6.4 hr. Approximately 50% of enrofloxacin is metabolised to the active substance ciprofloxacin. Ciprofloxacin is eliminated from the body at a half-life of 6.8 hr.

After intravenous injection at a dose rate of 5.0 mg enrofloxacin per kg body weight to lactating cows, peak plasma concentrations of approx. 23 mg/L are reached immediately. The overall drug exposure in plasma is 4.4 mg*hr/L. Enrofloxacin is eliminated from the body at a half-life of 0.9 hr. Approximately 50% of parent compound is metabolised to ciprofloxacin with peak plasma concentrations of 1.2 mg/L reached at 0.2 hr. Mean elimination half-life of ciprofloxacin is 2.1 hr.

In milk the metabolite ciprofloxacin mainly accounts for antibacterial activity (approx. 90%). Ciprofloxacin reaches peak milk concentrations of 4 mg/L within 2 hr after intravenous dosing. Total exposure in milk over 24 hours is approx. 21 mg*hr/L. Ciprofloxacin is eliminated from milk at a half-life of 2.4 hr. Peak concentrations of 1.2 mg enrofloxacin per litre are reached in milk within 0.5 hours with an total enrofloxacin exposure in milk of approx. 2.2 mg*hr/L. Enrofloxacin is eliminated from milk at 0.9 hr.

Pigs:

After intramuscular administration of 7.5 mg/kg body weight to pigs a mean peak serum concentration of 1.46 mg/L was achieved within 4 hours. The overall drug exposure over 24 hours was 20.9 mg*hr/L. The

drug was eliminated from the central compartment at a terminal half-life of 13.1 hr. With peak concentrations less than 0.06 mg/L mean serum concentrations of ciprofloxacin were very low.

Enrofloxacin has a high volume of distribution. The concentrations in the tissues and organs mostly significantly exceed serum levels. Organs in which high concentrations can be expected include the lungs, liver, kidneys, gut and muscle tissue.

Enrofloxacin is eliminated renally.

Indication

Cattle:

For the treatment of respiratory tract infections caused by enrofloxacin-sensitive *Histophilus somni*, *Mannheimia haemolytica*, *Pasteurella multocida* and *Mycoplasma* spp. For the treatment of mastitis caused by enrofloxacin-sensitive *E. coli*.

Pigs:

For the treatment of bacterial bronchopneumonia caused by enrofloxacin-sensitive *Actinobacillus pleuropneumoniae*, *Pasteurella multocida* and complicated by *Haemophilus parasuis* as a secondary pathogen in pigs.

Recommended Dose

Cattle:

The dosage for respiratory disease is 7.5mg Enrofloxacin per kg body weight for a single treatment by subcutaneous injection. This is equivalent to 7.5ml of the product per 100kg body weight per day. Do not administer more than 15ml (cattle) or 7.5ml (calf) per injection site. In case of serious or chronic respiratory disease a second injection may be required after 48 hours.

The dosage for the treatment of colimastitis is 5mg Enrofloxacin per kg body weight by intravenous injection. This is equivalent to 5ml of the product per 100kg body weight per day. The treatment of colimastitis should be exclusively by intravenous application on 2 to 3 consecutive days.

Pigs:

The dosage for respiratory tract infections is 7.5mg Enrofloxacin per kg body weight for a single treatment by intramuscular injection into the neck muscles behind the ear. This is equivalent to 0.75ml of the product per 10kg body weight per day. Do not administer more than 7.5ml per injection site. In cases of serious or chronic respiratory disease, a second injection may be required after 48 hours.

Method of administration: Repeated injections should be made at different injection sites. To ensure administration of the correct dosage, body weight should be determined as accurately as possible to avoid underdosing.

Route of Administration

To be given by subcutaneous or intravenous injection in cattle.

To be given by intramuscular injection in pigs.

Contraindication

Do not use for prophylaxis.

Do not use in cases of hypersensitivity to the active substance or to any of the excipients.

Do not use in known cases of resistance to other (fluoro)quinolones due to the potential for cross-resistance.

Do not use in animals with central nervous system-associated seizure disorders. Do not use in the presence of existing disorders of cartilage development or musculoskeletal damage around functionally significant or weight-bearing joints.

Warning and Precautions

Special warnings for each target species:

None.

Special precautions for use

Special precautions for use in animals:

For repeated injection or for injection volumes exceeding 15ml (cattle) or 7.5ml (pigs, calves) in divided doses, a new site must be chosen for each injection.

Official and local antimicrobial policies should be taken into account when the product is used.

Fluoroquinolones should be reserved for the treatment of clinical conditions which have responded poorly, or are expected to respond poorly, to other classes of antimicrobials.

Whenever possible, fluoroquinolones should only be used based on susceptibility testing.

Use of the product deviating from the instructions given may increase the prevalence of bacteria resistant to the fluoroquinolones and may decrease the effectiveness of treatment with other quinolones due to the potential for cross resistance.

Enrofloxacin is eliminated renally. As with all fluoroquinolones, delayed excretion can therefore be expected in the presence of existing renal damage.

Special precautions to be taken by the person administering the veterinary medicinal product to animals:

Direct contact with the skin should be avoided due to sensitisation, contact dermatitis and possible hypersensitivity reactions.

People with known hypersensitivity to (fluoro)quinolones should avoid contact with the product.

Wash hands after use.

In the event of accidental splash into the eye, rinse with large amounts of clean water. If irritation occurs, seek medical advice.

Do not eat, drink or smoke while handling the product.

Take care to avoid accidental self-injection.

In case of accidental self-injection, seek medical advice immediately and show the package leaflet or label to the physician.

Interaction with Other Medicaments

Antagonist effects due to concurrent administration of macrolides and tetracyclines may occur. Enrofloxacin may interfere with the metabolism of theophylline, decreasing theophylline clearance resulting in increased plasma levels of theophylline.

Pregnancy and Lactation

May be used during pregnancy and lactation.

Side Effects

In rare cases, transitory inflammatory reactions (swelling, redness) can occur at the injection site. These regress within a few days without further therapeutic measures. In rare cases, intravenous treatment can cause shock reactions in cattle, probably as a result of circulatory disturbances.

Gastrointestinal disturbances may occur in isolated cases during treatment of calves.

The frequency of adverse reactions is defined using the following convention:

- very common (more than 1 in 10 animals treated displaying adverse reaction(s))
- common (more than 1 but less than 10 animals in 100 animals treated)
- uncommon (more than 1 but less than 10 animals in 1,000 animals treated)
- rare (more than 1 but less than 10 animals in 10,000 animals treated)
- very rare (less than 1 animal in 10,000 animals treated, including isolated reports)

Symptoms and Treatment of Overdose

In cattle, a dose of 25mg/kg body weight administered by the subcutaneous route for 15 consecutive days is tolerated without any clinical symptoms. Higher doses in cattle and doses of around 25mg/kg and above in pigs may cause lethargy, lameness, ataxia, slight salivation and muscle tremors.

Do not exceed the recommended dose. In accidental overdose there is no antidote and treatment should be symptomatic.

Storage Condition

Store below 30°C. Protect from light.

Shelf life

3 years.

Following withdrawal of first dose, use the product within 24 hours. Any unused material should be discarded.

Withdrawal Periods

Cattle:

Subcutaneous injection:

Meat and offal: 14 days

Milk: 120 hours

Intravenous injection:

Meat and offal: 7 days

Milk: 72 hours

Pigs:

Meat and offal: 12 days

Packing

20ml, 50ml, 100ml, 250ml.

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	Manufacturer & Product Registration Holder
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